

UNSUNG WOMEN WARRIORS OF INDIAN SCIENCE

Asima Chatterjee

UNLOCKING NATURE'S CHEMISTRY



K. Venkataraman, Amritanshu Vajpayee,
Nandini Phanse, Shilpa Parikh, Brajesh Pandey

UNSUNG WOMEN WARRIORS OF INDIAN SCIENCE SERIES

ASIMA CHATTERJEE

UNLOCKING NATURE'S CHEMISTRY



(1917-2006)

National Pledge

India is my country.

All Indians are my brothers and sisters.

I love my country,
and I am proud of its rich and varied heritage.

I shall always strive to be worthy of it.

I shall give respect to my parents, teachers
and elders and treat everyone with courtesy.

To my country and my people, I pledge my
devotion.

In their well being and prosperity alone,
lies my happiness.

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Authors

K. Venkataraman

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Brajesh Pandey



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D-12, South Extension-I, New Delhi- 110049

E-mail: vijnanabharati@gmail.com

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ASIMA CHATTERJEE : UNLOCKING NATURE'S CHEMISTRY

Authors: K. Venkataraman, Amritanshu Vajpayee, Nandini Phanse,
Shilpa Parikh, Brajesh Pandey

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DEDICATED TO ALL THOSE IN THE
PURSUIT OF KNOWLEDGE.

चरन्मार्गान्विजानाति ।

A WANDERER (EVENTUALLY)
FINDS THE PATH.

**उत्तिष्ठत जाग्रत प्राप्य वरान्निबोधत ।
क्षुरस्य धारा निशिता दुरत्यया दुर्गमपथः ॥**

"Arise, awake, and learn by
approaching the noble ones.
The razor's edge is sharp and
hard to cross
— so is the path to wisdom."

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Foreword

Scientific biographies, when written with care for evidence and context, do more than just recount individual accomplishments. They help us understand how disciplines evolve and how institutions take shape. The present volume offers a thoughtful and well-structured account of the life and scientific contributions of Prof. Asima Chatterjee, situating her work within the formative years of modern chemical research in India. Drawing on her pioneering studies in natural products chemistry—especially the isolation and characterization of alkaloids and other bioactive compounds—the book brings out the intellectual discipline and sustained effort that marked her scientific journey.

The narrative follows her academic trajectory across important institutional settings, reflecting the research culture of her time. It presents, with clarity, her engagement with medicinal plant chemistry, where careful experimentation, chemical standardization, and reproducibility were not merely technical choices but guiding principles. Through this lens, the book shows how her work contributed to examining traditional knowledge systems with scientific rigour, thereby strengthening their relevance in modern inquiry.

Equally compelling is the attention given to her role as a mentor and institution-builder. Her academic leadership helped shape a generation of researchers who carried forward her emphasis on precision and disciplined thinking. Importantly, the account does not idealize her journey; instead, it acknowledges the constraints and challenges she navigated, offering a more grounded and authentic picture of scientific practice in mid-twentieth century India.

In its scope and treatment, this work will be of value to students, educators and scholars interested in the development of chemical sciences in India, as well as in the contributions of one of its foremost practitioners of the modern-day era in the form of Prof. Asima Chatterjee. By combining careful documentation with reflective analysis, the book presents a balanced and academically meaningful portrayal.

For participants of Vidyarthi Vigyan Manthan (VVM) and young learners, this narrative is an invitation to engage with science through attentive observation, disciplined inquiry and intellectual honesty. May it inspire you to pursue knowledge with both rigour and a clear sense of purpose.

Prof. Gobardhan Das
Member, Niti Aayog

Preface

"Two roads diverged in a wood, and I-
I took the one less traveled by,
And that has made all the difference."

- Robert Frost

The life of Professor Asima Chatterjee (née Mookerjee) (1917–2006) offers Indian students a rare opportunity to see how deep scientific achievement can grow from very humble beginnings. Born into an orthodox joint Hindu family in Kolkata, she entered a co-educational chemistry classroom at a time when many elders thought that higher education for girls was unnecessary, even dangerous. Her journey, from a young girl fighting for the right to study in Scottish Church College to the first woman to earn a D.Sc. from an Indian university, shows that scientific excellence is never created by talent alone; it is shaped by courage, persistence, and the support of a few determined people at crucial moments.

This book has been written with school and college students in mind, especially those who sometimes wonder whether science is “for people like us”. Through the story of Asima Chatterjee’s life, we hope to show that scientists are not distant geniuses who live only in laboratories; they are human beings who face family pressures, financial difficulties, institutional resistance, personal loss, and self-doubt, and yet continue to work because they care about knowledge and about people. For teachers, this biography offers material for classroom discussion in chemistry, biology, value education, history of science, and even civics and legal studies.

Traditional plant-based medicine is an important part of the Indian Knowledge System (IKS), reflecting integration of ancient wisdom with ongoing healthcare practices. The importance of traditional knowledge reflects from the fact that the Government of India established the Indian Knowledge Systems Division, in October 2020, under India's Ministry of Education, at AICTE, New Delhi, to promote, preserve, and disseminate the country's rich, traditional, and interdisciplinary knowledge. Among the pioneers who bridged tradition and modern science, Asima Chatterjee holds a place of distinction.

This book draws inspiration from her remarkable legacy, one that revived confidence in indigenous medicinal systems and paved the way for continued exploration of natural and integrative healthcare. Her contributions serve as a guiding light, reaffirming the enduring value of India's botanical wisdom and its relevance in shaping the future of medicine.

Professor Chatterjee's research field, natural products chemistry, may sound technical at first.

In simple terms, she devoted her life to understanding the chemicals present in Indian medicinal plants and to using that knowledge to develop safe and affordable medicines. Her work helped to clarify the structures of complex molecules such as indole alkaloids and coumarins, to discover new compounds, and to design drugs for epilepsy and malaria that could be produced from indigenous herbs without harmful side effects. Throughout the chapters, technical terms are introduced slowly and in context; where necessary, explanatory boxes give basic definitions so that students can follow the ideas without oversimplifying the science itself.

Equally important is another dimension of her life. As a teacher, department head, and institution builder at the University of Calcutta and beyond, Asima Chatterjee showed how universities can serve as the backbone of national scientific and technological training.

She did not see chemistry as a private career; she saw it as a social responsibility.

She worked to create research laboratories, to attract funding from national and international agencies, to fight legal battles on behalf of Indian companies, and to ensure that Ayurvedic knowledge and modern scientific methods could support one another.

This narrative draws upon multiple authentic sources: the detailed Biographical Memoir published by the Indian National Science Academy, Indian Academy of Science, contemporary tributes written by colleagues and students, institutional records of the University of Calcutta, and lists of her scientific publications and honours. At the same time, the book tries to bring out aspects of her personality that official documents cannot fully capture: her deep spiritual life, her loyalty to the ideals of Ramakrishna, Vivekananda, her refusal to accept royalties on drugs and books, her affection for students who still remember her simply as "Mastermoshai", "Didi", or "Ma".

We have tried to maintain a balance with clear and simple language. Single sentence paragraphs have been avoided so that the narrative flows in linked thoughts, as in a well-structured lecture or a carefully written letter.

From time to time, reflection prompts invite students to pause, think, and connect Asima Chatterjee's choices with their own lives and surroundings.

Remember, if institutions shape scientists, scientists also shape institutions. To understand how modern Indian medicinal chemistry matured, we must begin not with awards or laboratories, but with a classroom where intellectual curiosity first encountered disciplined method. If, after reading this biography, even one student decides to look more closely at the plants growing around a village pond, to ask what chemicals they might contain and how they might heal, or to persevere with an experiment that refuses to work, then the purpose of these pages will have been fulfilled. If some readers also come away with a deeper respect for universities, public institutions, and the quiet labour of teachers and researchers, they will be walking in the spirit of Professor Asima Chatterjee's own life.

K. Venkataraman
Amritanshu Vajpayee
Nandini Phanse
Shilpa Parikh
Brajesh Pandey

List of Abbreviations

AICTE	All India Council for Technical Education
B.Sc.	Bachelor of Science
Caltech	California Institute of Technology
CCRAS	Central Council for Research in Ayurvedic Sciences
COVID-19	Corona virus disease 2019
CSIR	Council of Scientific & Industrial Research
D.Sc.	Doctor of Science
DST	Department of Science & Technology
F.N.A.	Fellow of the Indian National Science Academy
FRS	Fellow of the Royal Society
IKS	Indian Knowledge Systems
INSA	Indian National Science Academy
ISc	Intermediate of Science
IUPAC	International Union of Pure and Applied Chemistry
M.Sc.	Master of Science
MBBS	Bachelor of Medicine and Bachelor of Surgery
MS	Master of Surgery
NMR	Nuclear Magnetic Resonance
Ph.D.	Doctor of Philosophy
SAR	Structure–Activity Relationship
U.S.A.	United States of America
UNESCO	United Nations Educational, Scientific and Cultural Organization
UV	Ultraviolet

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CHAPTER 1

Roots of Diligence: Childhood, Education, and the Making of a Scientist (1917–1947)

*"The child is father of the man."
William Wordsworth*

On a September morning in 1917, in the intellectually vibrant city of Calcutta (now Kolkata), a child was born into a household where books were deeply valued. That child, Asima Chatterjee, would grow into one of India's most distinguished chemists. Yet at the moment of her birth, neither the scientific community nor the nation, still under colonial rule, could have foreseen how profoundly her life would intersect with India's intellectual awakening. Her story begins not in a laboratory, but in a home where curiosity was nurtured with quiet conviction. Her life is the story encircling Calcutta as an intellectual centre, the culture of early twentieth century academia, and the unusual determination required for a woman to pursue doctoral chemistry at that time.

Early Years and Family Background

Asima Chatterjee was born (as Asima Mookerjee) on 23 September 1917 to Dr. Indra Narayan Mookerjee, a physician with a scholarly inclination towards Botany, and Kamala Devi, whose disciplined aura shaped the domestic environment in which young Asima grew with her younger brother, Sarashi Ranjan. Her father's profession ensured that scientific conversation was familiar in the household. Discussions on health, medicinal plants, and empirical reasoning flowed naturally across the dinner table. This early curiosity guided their careers, Asima became a pioneering chemist in medicinal plant research, while Sarashi Ranjan became a renowned surgeon and collaborated with his sister in her research on medicinal plants.

The Bengal of her childhood was alive with reform movements, literary renaissance, and nationalist aspiration. The atmosphere combined intellectual ferment with social transformation. Women's education, though still constrained by custom, was gaining legitimacy in progressive circles. It was within this transitional milieu that her intellectual character began to flourish.

From an early age, she exhibited an unusual attentiveness to patterns whether in language, arithmetic, music, or the natural world. She was interested in vocal music since her childhood and received training in classical music, *Dhrupad* and *Khayal*, for over fourteen years and stood second in the All-Bengal Music Competition in 1933. Her parents took special care to see that she was well conversant in Sanskrit which enabled her to read the great *epics* of the renowned writers of ancient India .

Family recollections later suggested that she was less interested in rote learning and more inclined to ask why things were the way they were. Such questioning, especially from a girl in the 1920s, was not always encouraged in wider society. Yet her family recognized in her not stubbornness but seriousness. They ensured that her schooling would not be compromised by social restrictions.

Early education: Matriculation and Intermediate Studies

Asima's formal education began at a time when institutions for women were still few. She entered Bethune Collegiate School, one of the pioneering institutions for women's education in India. Founded in the nineteenth century, Bethune Collegiate School carried a legacy of reformist courage. Within its classrooms, young Asima encountered teachers who treated intellectual ambition in girls not as novelty but as responsibility. Mathematics and science did not intimidate her; rather, they seemed to provide a grammar for understanding the world. Asima passed her Matriculation Examination in 1932, from Bethune Collegiate School, Calcutta, securing a Bengal Government Scholarship.

In 1934, Asima excelled in her Intermediate of Science (ISc) Examination, at Bethune College receiving scholarships in recognition of her outstanding academic performance. She obtained a Bengal Government Scholarship, Nawab Latiff and Father Lafnot Scholarships of the University of Calcutta and the Hemprova Bose Memorial Medal.

Higher Education

The decision to pursue higher education in science was, however, neither common nor socially acceptable. In colonial Bengal, the sciences, particularly chemistry, were regarded as rigorous disciplines suited primarily for

Devoted to Science: A Family's Legacy of Dedication to Knowledge

Asima Chatterjee was deeply influenced by her family's intellectual environment. Her father, Dr. Indra Narayan Mookerjee, a physician with a keen interest in botany, inspired her fascination with medicinal plants, while her mother, Kamala Devi, strongly supported her academic ambitions. Despite social opposition from her conservative family, Kamala Devi's determination and encouragement enabled Asima to pursue Chemistry Honours at the co-educational Scottish Church College, one of the few institutions open to women at the time,

Her brother, Sarashi Ranjan Mookerjee, MBBS, MS, Ph.D., became a distinguished surgeon and collaborated with her on research related to medicinal plants, gaining recognition for his work on diseases such as hypertension, hypothermia, and epilepsy.

Interestingly, both, Asima and Sarashi Ranjan, were the recipients of the coveted Shanti Swarup Bhatnagar award in their respective fields of Chemical Sciences and Medical Sciences !!

Asima's husband, Professor Baradananda Chatterjee, D.Sc. F.N.A., an eminent physical chemist, encouraged her to continue her academic journey with his progressive outlook. Without his constant inspiration, encouragement and co-operation it would have been impossible for Asima to dedicate herself to the cause of science.

The legacy of scientific excellence continued in her family. Her daughter Julie, grew up with the same enthusiasm for the field of organic chemistry and was a professor of Chemistry at the University of Calcutta. Julie married Professor Avijit Banerji, also from the Department of Chemistry at the same University. Both Julie and Avijit became respected organic chemists and professors at the University of Calcutta, with Julie later heading the Department of Chemistry. Asima's grandson, Aniruddha, also pursued a brilliant academic career in the Life Sciences.

men. Laboratories were spaces of chemicals, apparatus, and long hours of experimental uncertainty. For a young woman to aspire to such a path required not only academic competence but strong moral grit, qualities that Asima clearly possessed. She enrolled at Scottish Church College under the University of Calcutta, choosing Chemistry as her honours subject. The Scottish Church College was a co-education institution and being member of an orthodox, joint Hindu family, severe objections were raised by the elders to her joining the college. It was the courage and sheer determination of her mother, Kamala Devi, that enabled Asima to get admission there. She was the only woman student out of three admitted in the Chemistry Department of Scottish Church College to complete higher education. From there she completed her B. Sc. with honours in Chemistry with outstanding performance, in 1936, and received the Basanti Das Gold Medal for her outstanding performance.

Asima obtained M.Sc. degree in Organic Chemistry from the University of Calcutta in 1938, and received the Calcutta University Silver Medal for securing second rank and Jogmaya Devi Gold Medal. She was awarded the Nagarjuna Prize and a gold medal from the University of Calcutta in 1940 for the most outstanding research conducted in the Department of Chemistry. In 1942, she received the Premchand Roychand Studentship, and was also honoured with the Mouat Gold Medal, one of the university's most prestigious distinctions, in 1944. Organic chemistry appealed to her not merely as a collection of reactions but as a channel of transformation. The idea that invisible molecular rearrangements could alter colour, scent, medicinal potency, or toxicity fascinated her deeply. She would later describe organic chemistry as “the architecture of life rendered visible through reasoning.”

The intellectual environment of Calcutta during the 1930s was remarkably fertile. The city had already witnessed the pioneering work of Acharya Prafulla Chandra Ray in chemical education and industry, and the presence of scholars such as C. V. Raman had elevated Indian science to global recognition. Though she did not directly study under all these luminaries, their intellectual aura shaped the academic climate she inhabited.

During her post-graduate studies at the University College of Science, Calcutta, she came in close contact with eminent teachers and educationists like Acharya Prafulla Chandra Ray, Professors Prafulla Chandra Mitter, Pulin Behari Sarkar, Jogendra Chandra Bardhan, Prafulla Kumar Bose, Jnanendra Nath Mukherjee and Priyadarshan Ray, all of whom influenced her future career.

Commitment to Academic Teaching and Mentorship

In 1940 Asima joined Lady Brabourne College, Calcutta, as the founder, Head of the Department of Chemistry. In 1944 she was appointed a Honorary Lecturer in Chemistry, at the University of Calcutta. Upon resuming these responsibilities, she confronted realities that would have discouraged a less determined scholar. Scholarships were rare and insufficient. Students worked in shifts because laboratory space was limited. Access to modern instrumentation was minimal. Yet she did not lament these constraints. Instead, she created opportunities. Research under her supervision became synonymous with intellectual exactitude. Students learned that limited resources demanded sharper thinking. If an instrument was unavailable, reasoning had to be sharper; if chemicals were expensive, experiments had to be carefully designed before execution.

From Research to Recognition: Asima's D. Sc. Achievement in 1944

In the laboratories of colonial India, where few women ventured into scientific research, Asima began her journey, quietly shaping a legacy that would illuminate the world of medicinal chemistry. Her scientific career began in the early 1940s, when her deep interest in organic chemistry transformed into pioneering research on plant based compounds, laying the foundation for modern natural products chemistry in India.

Asima began her postgraduate research at the University of Calcutta, under the guidance of Professor Prafulla Kumar Bose, a reputed chemist and pioneer in natural product chemistry. It was under his supervision that she encountered the intricate world of plant alkaloids, the complex organic molecules often responsible for the pharmacological properties of medicinal plants. The choice of this field would define her scientific trajectory for decades to come. Natural product chemistry demanded patience and precision. Unlike synthetic chemistry, where reactants were defined and controlled, plant chemistry required extraction, separation, purification, and structural deduction from mixtures of unknown complexity.

Her doctoral research focused on the chemistry of plant alkaloids and coumarins, compounds with significant pharmacological relevance. The technical aspects of her work involved isolating chemical constituents from plant material and determining their structural characteristics through classical methods of analysis. At a time when sophisticated spectroscopic tools were limited, structural elucidation relied heavily on systematic chemical transformations and logical inference. Each structural hypothesis had to

What is natural product chemistry and phyto-chemistry?

Natural products are substances found in nature, originating from plants, animals, and microorganisms. Natural product chemistry is defined as the study and utilization of chemical compounds derived from natural sources. Historically, natural plant products have played an important role in the development of pharmaceutical drugs for a number of diseases including cancer and infection.

Phyto-chemistry is the study of natural chemicals derived from plants, known as phyto-chemicals. Phyto chemists study structures and functions of the large number of secondary metabolites found in plants, which helps in studying the functions of these compounds in human and plant biology, and the biosynthesis of these compounds. Phyto-chemicals are bioactive, non-nutrient compounds produced by plants to protect themselves from threats like pathogens and UV radiation. When consumed, they provide potential health benefits to humans, such as reducing the risk of chronic diseases (cancer, cardiovascular disease), providing antioxidant protection, and lowering inflammation. The compounds found in plants are of many kinds, but most can be grouped into four major biosynthetic classes: alkaloids, phenylpropanoids, polyketides, and terpenoids.

withstand experimental testing. The discipline this required shaped her scientific temperament permanently.

The early 1940s were a turbulent period in India. The Second World War disrupted supply chains, laboratory access, and academic routine. Simultaneously, the nationalist movement intensified. Universities were spaces not only of scholarship but of political awareness. Amid such instability, pursuing sustained research required resilience. Asima's doctoral work progressed steadily despite these disruptions. In 1944, she earned the Doctor of Science (D.Sc.) degree from the University of Calcutta, for her research on *'Naturally Occurring Indole Alkaloids and Coumarins'*, becoming the first Indian woman to receive a doctorate in science from an *Indian* university. Likewise, as should be noted here, earlier in 1931, Janaki Ammal broke barriers by becoming one of the first Indian women to earn a D.Sc. from a *foreign university*, the University of Michigan in the USA, further illustrating how determination and resilience can overcome societal limitations and pave the way for future generations in science.

The Intellectual Grammar of Natural Product Chemistry

Asima's research on alkaloids, particularly indole alkaloids, demanded not only patience but a refined logical imagination. Alkaloids are nitrogen containing organic compounds, frequently physiologically active, and often structurally intricate. Many contain fused ring systems, stereochemical complexity, and subtle functional group arrangements that influence biological activity. Their isolation from plant matrices is an exercise in separation science; their structural determination is an exercise in deductive reasoning.

Think Like a Chemist

If two molecules differ only by the position of a single functional group yet display different biological effects, what does that reveal about molecular recognition in the body? The study of such differences forms the foundation of rational drug design.

In the mid-twentieth century, modern spectroscopic conveniences such as high-resolution NMR spectroscopy, advanced mass spectrometry, and X-ray crystallography were either unavailable or only partially accessible in Indian laboratories. Structural elucidation depended heavily upon classical methods: elemental analysis to determine molecular formulae, measurement of melting points to infer purity, optical rotation to detect chirality, and

Chirality (or Chiral)

Chirality (or chiral) refers to something that cannot be superimposed on its mirror image, similar to how a right hand cannot fit perfectly into a left-handed glove. These molecules exist as mirror-image pairs, called enantiomers.

- A molecule is chiral if it has:
- A central carbon (called a chiral center)

Four different atoms or groups attached to it

This "handedness" is crucial in chemistry, as chiral molecules (enantiomers) often exhibit different biological activities, making them highly significant in pharmaceuticals.

chemical degradation reactions to fragment molecules into identifiable subunits.

To illustrate the intellectual discipline involved, consider the process of deducing an alkaloid structure. A plant extract would first be subjected to acid-base extraction, exploiting the basic nature of alkaloids to separate them from neutral compounds. Subsequent fractional crystallization or chromatography would yield purified samples. The chemist would then determine elemental composition and molecular weight. From there, a series of carefully designed reactions like oxidation, reduction, hydrolysis, acetylation, would generate derivatives whose properties could be compared. Each reaction functioned like a question posed to the molecule: What functional group do you contain? How many rings form your backbone? Where is the nitrogen positioned? The molecule did not answer verbally; it responded through changes in melting point, solubility, or reactivity.

Try to Imagine

Imagine attempting to reconstruct a three-dimensional architectural structure without being allowed to see it directly. You are permitted only to remove small sections, test how they behave, and infer the whole from fragments. That was the daily intellectual exercise of structural chemists in the era of Prof. Asima Chatterjee.

Such work required extraordinary patience. Weeks of labour could culminate in a single structural correction. Yet this painstaking methodology shaped a generation of chemists who understood that certainty in science is earned, not assumed.

Marital Life and Its Impact on Asima's Career

The intellectual rigor of Asima's research years coincided with personal developments. In 1945 she married Dr. Baradananda Chatterjee, a fellow academian who shared her respect for research. Professor Baradananda Chatterjee, D.Sc. F.N.A. (Fellow of the Indian National Science Academy), was a renowned physical chemist who became Professor and Head of the Department of Chemistry, Geology and Metallurgy and Vice-Principal, Bengal Engineering College, Howrah. Professor Baradananda Chatterjee, had a profound influence on his wife. Without his constant inspiration,



(Figure 1.1: Prof. Asima Chatterjee with husband Prof. Baradananda Chatterjee, and daughter Julie in the USA. Rights: Prof. Julie Banerji, personal family archives. Credit: Indian Academy of Science.)

encouragement and co-operation it would have impossible for Asima Chatterjee to dedicate herself to the cause of Science. They had a daughter, Julie, their only child, who later became a professor of organic chemistry at Calcutta University.

Prof. Asima's Chatterjee's life had a unique example of harmony and commitment between professional and home-life. She had always been a devoted wife, mother, daughter, elder sister, mother-in-law and grandmother. She would rise up early in the morning and would complete all household chores, including cooking, before she left for the University. On returning, late in the evening, she would again return to house-work.

Marriage did not diminish her research ambitions; rather, it coexisted with them. Their partnership was characterized by mutual regard for intellectual work. Navigating professional and domestic roles in mid-twentieth century India required negotiation of social expectations. She managed these transitions not through confrontation but through consistent excellence, allowing achievement itself to argue her case.

The academic mindset and open- minded nature of her husband, Baradananda Chatterjee, motivated Asima Chatterjee to pursue higher

education after her doctoral research. She left for the USA in 1947 accompanied by her eleven-month-old daughter and a governess. This was an extraordinary step at a time when India was still under British rule, a young, middle-class woman from a conservative Bengali family demonstrated remarkable courage and determination by going abroad to pursue higher education with her infant child.

Chapter 2

Shaping a School of Natural Product Chemistry: Vision, Dedication and Nation Building (1947 –1975)

"To strive, to seek, to find, and not to yield."

Alfred Tennyson

International Exposure, Indigenous Commitment: Expanding Scientific Horizons

The period immediately following her doctoral degree broadened her scientific horizons. In 1947 she left for the U.S.A. on study leave from Lady Brabourne College. Asima Chatterjee undertook advanced research abroad, including work at the University of Wisconsin and later at the California Institute of Technology (Caltech) and the University of Zurich, Switzerland.

In the US, she worked with L. M. Parks at the University of Wisconsin, Madison, on naturally occurring glycosides and with L. Zechmeister, at the California Institute of Technology, Pasadena, USA, on carotenoids and provitamins. She was awarded the coveted Watumull Fellowship in recognition of her research work in the USA.

Asima Chatterjee moved to University of Zurich in 1949. In Europe, she worked with the 1937 Chemistry Nobel Prize winner Paul Karrer at the University of Zurich, on biologically active alkaloids. She was so fascinated while working with biologically active alkaloids that she dedicated the rest of her research career to the field of natural product chemistry. These international exposures introduced her to contemporary methods and global research networks. Yet even during these engagements, her scientific curiosity



(Figure 2.1: Prof. Asima Chatterjee at Zurich University 1949, Rights: Prof. Julie Banerji, personal family archives. Credit: Indian Academy of Sciences.)

remained anchored in Indian medicinal plants. She did not perceive Western laboratories as destinations of intellectual migration but as sites of exchange. Her intent was always to return with sharpened tools and broader perspective.

When Asima Chatterjee returned to India from the USA, in 1950, she did not return merely as a trained chemist; she returned as a scientist with a vision. The country had recently attained independence, and its intellectual infrastructure was still fragile. Laboratories were modest, equipment was scarce, research funding was uncertain, and yet the air carried an unmistakable optimism. For Asima Chatterjee, this expectation was not abstract policy, it was a personal commitment.

Her return marked the beginning of an extraordinary period in which she would not only deepen her own research in natural product chemistry but also create an institutional culture that transformed the Department of Chemistry at the University of Calcutta into an internationally respected centre of excellence. After her return to India from the U.S.A. in 1950, she started research on alkaloids and coumarins with renewed vigour.

Indole Alkaloids and the *Rauwolfia* Investigations

One of the central themes of her research was the study of indole alkaloids derived from species of *Rauwolfia*. These plants had long been used in traditional medicine, particularly for hypertension and neurological disorders. The discovery of reserpine earlier in the twentieth century had already demonstrated the profound therapeutic potential of such compounds. However, the broader chemical family of related alkaloids remained insufficiently characterized.

Her work on *Rauwolfia* species brought her into close association with the late Professor Dr. Salimuzzaman Siddiqui, FRS, former Director of Husein Ebrahim Jamal Post Graduate Institute of Chemistry, University of Karachi, Pakistan.

Her work contributed significantly to clarifying the structural relationships among various indole alkaloids. These compounds typically feature an indole nucleus, a bicyclic structure consisting of a benzene ring fused to a pyrrole ring, integrated into more elaborate ring systems. Small variations in substitution patterns or stereochemistry can drastically alter biological properties. Determining these subtle differences required analytical precision. Through systematic degradation studies and comparative analysis, her laboratory helped resolve structural ambiguities in several related alkaloids. In doing so, she contributed to global understanding of indole alkaloid biosynthetic patterns. The ability to recognize structural families within plant species also provided insight into evolutionary and biochemical relationships among plants.

Her review articles in international volumes on alkaloid chemistry did more than summarize findings; they synthesized patterns across multiple studies. By organizing structural data into coherent classifications, she helped transform scattered discoveries into an interpretable framework. Such interpretative work is often less visible than primary discovery, yet it is crucial for scientific progress.

Coumarins and Structure–Activity Relationships

In addition to alkaloids, she investigated coumarins and related compounds. Coumarins are benzopyrone derivatives found widely in plants and known for diverse biological activities. Studying them required attention not only to structure but also to structure–activity relationships. How does substitution at a particular carbon atom alter pharmacological potency? Does hydroxylation increase solubility? Do methyl groups alter metabolic stability?

The concept of structure–activity relationship (SAR) was becoming increasingly central to medicinal chemistry during her career. SAR analysis rests on the principle that molecular structure determines biological interaction. By comparing related compounds and observing variations in activity, chemists infer which structural features are essential. Her work on plant-derived compounds anticipated later systematic SAR methodologies in pharmaceutical chemistry. It is important to recognize that the scientific identity forming during these years was neither technical nor theoretical. She understood, even early on, that India's botanical heritage represented a reservoir of chemical knowledge awaiting systematic exploration. Traditional medicinal practices were not, in her view, relics of superstition; they were empirical traditions requiring scientific articulation. This conviction would later guide her most significant contributions.

As 1947 approached and India stood on the threshold of independence, Asima Chatterjee had already established herself as a scholar of rare determination. The nation was poised to define its intellectual sovereignty. Universities were to become engines of development. Scientific research was to serve public health and industrial progress. For a chemist trained in the structural secrets of medicinal plants, the moment carried particular resonance.

The chapter of her life that began in 1917 had instilled discipline, resilience, and vision. She had crossed thresholds of gender expectation, academic hierarchy, and colonial limitation. Yet her journey was only beginning. The laboratory awaited expansion. Students would soon gather around her. And the molecules she had learned to decipher would lead her toward questions far larger than structural formulas. The doctoral thesis did not conclude a journey; it initiated one. Having acquired the tools of structural reasoning, she now faced the larger question: how might organic chemistry respond to the medicinal challenges of a newly independent nation?

In the years to come, she would not merely practice chemistry; she would build an institution, mentor generations, and transform natural product research into a national endeavour. The roots had been laid deep. The next chapter would see those roots give rise to a flourishing scientific school.

The Scientific Journey Ahead

Asima's appointment as Reader in the Department of Chemistry at the University of Calcutta in 1954 marked a turning point. At that time, there was only one such post in the department. The recognition was not ceremonial.

***Rauwolfia*: A Medicinal Treasure of Nature**

Rauwolfia is a genus of medicinal plants in the botanical family *Apocynaceae*. These plants are well known for their pharmacologically important alkaloids, especially those used in treating hypertension and mental disorders. The genus is named to honour Leonhard Rauwolf, a German physician and botanist.

Several species of the *Rauwolfia* genus have been studied. *R. serpentina*, *R. canescens*, *R. tetraphylla*, *R. micrantha*, *R. vomitoria* are some of the important species of *Rauwolfia*.

Amongst *Rauwolfia* species, *Rauwolfia serpentina* (Sarpa gandha), is a medicinally important herb, since ancient times. The shrub smells like a serpent and in Sanskrit, is named accordingly as *sarpa gandha*. The alkaloid *reserpine* is isolated from the root and is widely used to treat hypertension, mental agitation and other problems. The reserpine content in the root of *R. serpentina* has attracted worldwide attention for drug development.

A detailed investigation into *Rauwolfia canescens* was initiated by Asmia Chatterji (nee Mookerjee), in 1941. She studied the chemical structure, stereochemistry and pharmacological aspects of *rauwolscine*, the major alkaloid of *Rauwolfia canescens*. She also reported that the hypnotic alkaloid reserpine, first isolated in pure and crystalline state from *R. serpentina*, is found to occur in a number of *Rauwolfia* species.

It signalled confidence in her growing research leadership. By then, her work on indole alkaloids, particularly from species of *Rauwolfia*, had begun to attract international attention. She combined botanical knowledge, organic structural analysis, and mechanistic reasoning in ways that were both rigorous and original.

The choice of *Rauwolfia* species as a research focus was not accidental. The plant had long been used in traditional Indian medicine, and the discovery of reserpine as a therapeutic agent for hypertension had stimulated global scientific interest. Yet many structural questions remained unresolved, and new alkaloids awaited identification. Asima Chatterjee's laboratory became one of the most productive centres for isolating and elucidating the structures of these complex molecules. The work demanded not only chemical insight but patience. Plant material had to be collected, dried, extracted, fractionated,

purified and repeatedly analyzed before a single structural conclusion could be drawn.

To understand the magnitude of this task, one must remember that modern spectroscopic conveniences were limited in mid-twentieth century India. Nuclear magnetic resonance and high-resolution mass spectrometry were not readily available. Structural elucidation relied heavily on chemical degradation, derivatization, melting points, optical rotation, and painstaking logical deduction. Students trained under her developed a habit of drawing possible molecular frameworks repeatedly, erasing and redrawing until consistency emerged between experimental observation and theoretical reasoning. During these years, her reputation as a teacher deepened alongside her scientific achievements. Those who attended her lectures often recalled that she did not merely describe reactions; she narrated the inner logic of molecules. She would begin with a question- why does this functional group behave in a particular manner? - and gradually unfold the answer through stepwise reasoning, guiding students to think rather than memorize. The classroom became an extension of the laboratory.

From Appointment to Legacy: Transforming Natural Product Chemistry

In 1962 she was appointed the Kumar Guruprasad Singh Khaira Professor of Chemistry, a distinguished chair that she would hold until her retirement in 1982. The chair brought prestige, but more importantly, it expanded her influence over curriculum and research direction. She advocated strongly for strengthening natural product chemistry as a discipline rooted in India's botanical wealth. Her conviction was clear: the subcontinent's biodiversity represented not only ecological richness but chemical possibility.

As her laboratory grew, so did the diversity of research themes. Beyond *Rauwolfia*, her group investigated alkaloids from *Alstonia*, *Rhazya*, *Kopsia*, and *Vinca*. Each plant offered new structural puzzles. Among her notable contributions were the elucidation of structures and stereochemistry of compounds such as rauwolscine and studies that clarified aspects of ajmaline and sarpagine chemistry. These investigations did not occur in isolation; they connected Indian research to global debates in alkaloid chemistry. Reviews authored by her in respected international series demonstrated that she was not merely contributing data but shaping interpretation.

Pioneer in cancer research: Serving humanity through science

Prof. Asima Chatterjee was a pioneer in medicinal chemistry and cancer drug research, dedicating her life to discovering treatments derived from Indian medicinal plants. At university she majored in organic chemistry, specializing in research on *vinca alkaloids*. During the four decades of research on the chemical compounds, she also studied cancer and anti-cancer growth drugs, investigating how alkaloids could be used in chemotherapy. One breakthrough came with her work on *vinca alkaloids* from the Madagascar periwinkle plant, *Catharanthus roseus*, (synonymous with *Vinca rosea*, formerly included in the genus *Vinca* as *Vinca rosea*), which can help slow down some cancer cells by preventing them from duplicating. *Vinca alkaloids* (vincristine, vinblastine) are anti-mitotic and they interfere with microtubules, the cell structures that help move chromosomes during mitosis, and are used to treat cancer.

Alkaloid Chemistry and Reaction Mechanisms

Asima Chatterji's pioneering work on the alkaloids of *Rauwolfia*, *Alstonia*, *Kopsia*, *Rhazya* and *Vinca*, has made immense impact on researches that followed in the field of indole alkaloids, particularly novel 5 α -pregnane derivatives derived from the Apocynaceae and Buxaceae families. In terpenoids, she provided cutting-edge methods of transformation and correlation of their different skeletal patterns, notably through catalytic rearrangements and synthetic approaches.

Coumarins found in plant families such as Rutaceae, Umbelliferae, Compositae and Euphorbiaceae families, are yet another group of natural products which bear the imprint of her contributions. Additionally, her work in reaction mechanisms, especially the acid-catalyzed hydramine fission reaction, has been widely recognized and even cited in standard organic chemistry texts like I. L. Finar.

An Era of Global Admiration

A global admiration came calling for Prof. Asima Chatterjee in the year 1967 when the internationally acclaimed scientist Linus Pauling paid a visit to Prof. Asima Chatterjee during his tour to India. Asima Chatterjee's only meeting



(Figure 2.2: Prof. Asima Chatterjee (front row, third from right) with Prof. Linus Pauling, NL and her Research Scholars, at the Science College Campus, Kolkata in February 1967. Rights: Prof. Julie Banerji, personal family archives. Credit: Indian Academy of Sciences.)

with Linus and Ava Helen Pauling occurred during their visit to India in January–February 1967. In the final phase of their tour, the Paulings spent a brief time in Kolkata, during which they visited the university and toured Chatterjee’s laboratories.

Resilience Through Grief: Rising After Personal Tragedy

The decade of the 1960s also brought profound personal sorrow. In 1967 she lost both her father and her husband within a short span of time. The double bereavement was devastating, and she suffered a severe heart attack. For months she hovered between fragility and recovery. Yet what followed revealed the resilience that defined her character. It was through the influence and affection of Late Swami Abhayanandaji Maharaj of the Ramakrishna Math and Mission, Belur, that she regained her mental strength. Supported by spiritual guidance from the Ramakrishna Math and by the affection of her students and colleagues, she returned to work. Her recovery was not dramatic; it was gradual, and quiet. Science, for her, was not escape from grief but restoration of purpose.

As a mark of her deep regard and devotion to her late husband, Professor Baradananda Chatterjee, she had instituted the Baradananda Memorial Lectureship’ at the Bengal Engineering College, Howrah, to be delivered

Asima Chatterjee's Role in a Landmark Legal Case: A Scientist's Stand for Justice

One episode from 1968 deserves particular attention, for it reveals a dimension of Asima's scientific integrity beyond the laboratory. During a landmark legal dispute, in 1968, between Bengal Chemical and Pharmaceutical Works Ltd., founded by Acharya P.C. Roy, and Hoechst Co. Ltd. over the infringement of a patent involving a sulphonamide derivative, Asima Chatterjee agreed to serve as the principal witness for the Indian company on the condition that she would accept no payment. Asima's profound knowledge of Organic Chemistry, courage and conviction helped Bengal Chemical Pharmaceutical Works Ltd. win the legal battle.

In spite of her mental condition, due to the demise of her father and husband, in 1967, she answered hundreds of questions in Chemistry for days together in the Calcutta High Court. Had the verdict gone against the Bengal Chemical and Pharmaceutical Works Ltd., it would have faced liquidation.

every year by a leading chemist of India having significant contribution in the field of soil chemistry or corrosion.

Building Excellence: Leadership, Research, and Global Engagement:

By 1969 she became Head of the Department of Chemistry at the University of Calcutta, a position she held until 1979. Under her leadership, the department underwent significant transformation. She secured funds from the University Grants Commission to expand laboratory infrastructure. The construction of additional floors in the N.R. Sen Building allowed installation of sophisticated instruments and expansion of research laboratories. In 1972 the University Grants Commission sanctioned a Programme of Special Assistance for intensifying teaching and research in natural product chemistry, with her as Honorary Programme Coordinator. This recognition formalized what had already become evident: the department had emerged as a national centre in the field.

International engagement remained integral to her vision. She delivered lectures and chaired sessions at major symposia in Europe and Asia, including meetings organized under UNESCO and IUPAC. These visits were not

academic tourism; they were exchanges that kept her laboratory intellectually current. Students sensed that they were participating in a global conversation even while working in Kolkata.

Professor Chatterjee acted as Chairperson in the UNESCO symposia on Phytochemistry held in Kuala Lumpur (1957) and Hong Kong (1961), in the UPAC Symposia on Chemistry of Natural Products held in Zurich (1955), Australia (1960), Japan (1964) and JSSR (1970). She participated in the meetings of the British Association for the Advancement of Science in 1970 and 1971 and visited various British Universities.

Asima visited several countries on invitation to participate in scientific symposia and conferences and to deliver lectures on her research contributions. She delivered a number of *Oration* and *Convocation* lectures in Universities and Institutes throughout India. A large number of students obtained their Ph.D. and D.Sc. degrees under her guidance, many of whom are occupying topmost positions in academia and industry in India and abroad. Many of them have developed their Schools on Natural Product Chemistry and are playing key-roles in the development of the area in India and abroad, in colleges, research institutions, universities, industries and policy-making bodies.

Honours Earned, Horizons Expanded

In 1975, Prof. Asima Chatterji achieved two milestones that symbolized her stature. She was elected General President of the 62nd Session of the Indian Science Congress, becoming the first woman scientist to hold this position. In her presidential address, she emphasized the central role of universities in scientific and technological progress, arguing that they must receive national priority. The same year she was awarded the Padma Bhushan by the Government of India, recognizing not only her research contributions but her leadership in strengthening Indian science.

By the mid-1970s, Asima Chatterjee had accomplished more than personal distinction. She had built a school; a community of chemists trained in structural reasoning, respect for indigenous knowledge, and disciplined experimentation. Her laboratory had become a place where students did not merely complete theses but absorbed a scientific ethic.

As this chapter draws to a close, one senses that her work was expanding from structural chemistry toward application. The compounds isolated and analyzed in glass flasks were beginning to suggest therapeutic possibilities. Her longstanding interest in medicinal plants was gradually converging with

Historic 'Firsts'

- Asima Chatterjee was the first woman to be awarded a Doctor of Science by an *Indian* University, in 1944, by the University of Calcutta.
- First female recipient of the prestigious Shanti Swarup Bhatnagar Prize for Science and Technology in Chemical Science in the year 1961, for her valuable contributions in the field of Chemistry of Natural Products.
- She became the first Indian woman scientist to hold a Chair of any university in India, as the Khaira Professor of Chemistry at the University of Calcutta (1962-1982). Khaira Professor in Chemistry is one of the coveted Chairs of Excellence of the University of Calcutta.
- First Indian woman scientist elected as General President of the Indian Science Congress, a premier institution that oversees scientific research, in 1975.



(Figure 2.3: Prof. Asima Chatterjee during the felicitation ceremony at the University of Calcutta after receiving the Padma Bhushan Award in 1975. Rights: Prof. Julie Banerji, personal family archives. Credit: Indian Academy of Sciences.)

a broader public health vision. The next phase of her life would carry natural product chemistry from structural elucidation to drug development, bridging laboratory science and clinical hope.

The river of research that began in the classrooms of Bethune School and Scottish Church College had now gathered force. It was moving toward new terrain- where chemistry would meet medicine in ways that would affect lives beyond academia. A quiet spark in the realm of molecules and medicinal plants, the journey of Asima Chatterjee, thus destined to transform Indian science.

Chapter 3

From Molecules to Medicines: Bridging Research and Public Health (1975–1990s)

*"I slept and dreamt that life was joy;
I awoke and saw that life was service;
I acted and behold, service was joy."*

Rabindranath Tagore

By the mid-1970s, Asima Chatterjee had already secured her place among India's foremost organic chemists. She had built a reputed research school, mentored doctoral scholars, chaired national scientific forums, and strengthened institutional infrastructure. Yet a subtle transformation was underway in her intellectual orientation. Structural elucidation, though foundational, was no longer her sole horizon. Increasingly, her work began to turn toward a deeper question: how might the molecular insights painstakingly derived in laboratories translate into therapeutic benefit for society?

The transition was neither abrupt nor rhetorical. It emerged organically from decades of engagement with medicinal plants. Natural product chemistry, as she practiced it, had always stood at the interface of traditional knowledge and modern analytical method. The isolation of alkaloids and coumarins had revealed the chemical logic underlying ancient remedies. But structural knowledge alone was incomplete unless it could illuminate pharmacological potential. In this phase of her career, chemistry moved closer to medicine. Amongst the most significant directions of her later research was her sustained work on anti-malarial agents derived from indigenous plants. Malaria had long been one of India's most persistent public health challenges. The global reliance on quinine and later synthetic antimalarials had not



*(Figure 3.1: Prof. Asima Chatterjee delivering the presidential address at 62nd Indian Science Congress 1975. Rights: Prof. Julie Banerji, personal family archives.
Credit: Indian Academy of Sciences.)*

eliminated the disease, and drug resistance was already emerging as a serious concern. In this context, her attention to plant derived compounds was both scientifically and socially responsive.

From Isolation to Formulation: The Development of Anti-Malarial Research

The involvement of Prof. Asima Chatterjee in anti-malarial research represents a significant expansion from structural chemistry to applied medicinal science. Malaria, caused by *Plasmodium* parasites transmitted through mosquitoes, has a complex life cycle and remains difficult to eradicate. Drug resistance complicates treatment strategies. Investigating plant derived compounds offered a complementary pathway to synthetic drug development.

Working with her research group, she investigated bioactive compounds from several plant species, studying their chemical structures, reactivity, and therapeutic promise. Her contributions played an important role in the development of anti-malarial formulations such as Ayush-64, a polyherbal preparation rooted in Ayurvedic tradition but refined through modern pharmacological evaluation. The project exemplified her long-standing conviction that traditional medical knowledge should not be dismissed

Medicinal Plants in Indian Tradition: From Cultural Heritage to Contemporary Medicine

Traditional Indian medicine, especially systems like Ayurveda, depends greatly on the use of plants for healing and maintaining health. It is an important and enduring part of India's traditional medical heritage.

Knowledge about medicinal plants has been passed down through generations via texts, oral traditions, and community practices. Even today, traditional plant-based remedies remain highly relevant, and modern research continues to validate their effectiveness, showing how IKS contributes to both cultural heritage and contemporary healthcare. Today, Indian medicinal plants continue to bridge the cultural heritage and ancient wisdom with contemporary healthcare.

Traditional Indian medicine using plants is a vital part of Indian Knowledge System (IKS), which integrates ancient wisdom with daily life and healthcare practices. For many centuries, plants had been used for their medicinal properties. However, the biological mechanisms underpinning these plant's effects came to limelight only after the investigations of researchers like Dr. Asima Chatterji.

In modern times, the relevance of plant-based traditional medicine has grown globally. Scientific studies and contributions from researchers like Prof. Asima Chatterjee have helped validate the medicinal value of many Indian plants. Through her scientific work, she reminded the world about the glorious days of traditional Indian medicine using plants. She researched medicinal chemistry and developed leading anti-convulsive, anti-malarial, and chemotherapy drugs from natural products.

nor uncritically accepted; rather, it should be examined through rigorous chemical and biological analysis. Ayush-64 eventually gained recognition for its potential in malaria management and symbolized a rare synthesis between classical Indian medicine and laboratory science.

To appreciate the intellectual discipline involved in such work, one must understand that translating plant extracts into standardized formulations demands careful chemical characterization. Active constituents must be identified; dosage consistency must be established; toxicity profiles must be assessed; interactions must be understood. For a chemist trained in structural

Healing from Nature

Ayush-56 and Ayush-64

India has a rich tradition of using medicinal plants for healing, which has been strengthened by the contributions of eminent scientists like Asima Chatterjee. A pioneering organic chemist, she dedicated her life to studying plant-based compounds and developing effective medicines. Her work successfully bridged traditional knowledge and modern science, contributing to development and phytochemical standardization of important formulations such as *Ayush -64*, an anti-malarial drug and *Ayush -56*, used in the treatment of epilepsy. These formulations were patented by the Indian government.

Ayush-64

The anti-malarial drug, coded *Ayush -64*, which is a combination of different parts of four herbs was developed by Professor Chatterjee (Indian Patent No. 568/Del. 70, 7th August 1979). It is a polyherbal, non toxic, anti-malarial drug consisting extracts of: *Alstonia sholaris* (Saptaparna), *Picrorhiza kurroa* (Katuki), *Swertia chirata* (Chirayata) and *Caesalpinia crista* (Kuberaksha) This drug was repurposed for COVID,19 as its ingredients showed notable antiviral, immune modulator and antipyretic properties.

Ayush-56

The anti-epileptic drug *Ayush-56* developed by Professor Chatterji is used commercially. It consists of plant extracts of *Marsilea minuta* (Sunisannaka) and *Nardostachys jatamansi* (Jatamansi). She isolated and established the structure of marsilin, the sedative and anticonvulsant drug, from *Marsilea minuta* (commonly called as water fern). Marsilin has been found to be effective in the treatment of epilepsy and is used as a highly successful rehabilitation drug in combination with *Nardostachys jatamansi* under the code-name Ayush 56 (Indian Patent No. 141170 dt. 14th July 1976).

Both these combination-drugs have been patented by the Central Council of Research in Ayurveda and Siddha (CCRAS), under the Ministry of Health and Family Welfare, Government of India. The patents have been purchased by the National Research Development Corporation, Government of India, and the drugs are being marketed by several companies.

reasoning, this process was not a departure from earlier work but its logical extension. The same meticulous logic used to deduce the arrangement of atoms in alkaloids was now directed toward ensuring therapeutic reliability.

Try to Imagine

Imagine holding a dried fragment of medicinal bark in your hand. To the untrained eye, it is little more than plant matter. To a chemist like Asima Chatterjee, it represents a complex mixture of organic molecules, each with a distinct structure, each potentially interacting with the human body in specific ways. The task is to separate, identify, test, and understand these molecules. In that process, centuries of traditional observation meet the precision of modern science.

Science with Purpose: Developing Drugs for Human Well-Being

Prof. Asima Chatterji's work also expanded into anti-epileptic research, exploring plant-derived compounds that could offer alternatives or complements to existing medications. She developed Ayush -56, an anti-epileptic drug, consisting plant extracts of *Marsilea minuta* (water fern, Sunisannaka) and *Nardostachys jatamansi* (Jatamansi). While some of these investigations remained within the domain of exploratory research, they reflected a consistent orientation toward therapeutic relevance. Chemistry, for her, was never isolated from human consequence. For her, every discovery or experiment had real-life implications for human health and well-being. Her work on medicinal plants, anti-epileptic drug like Ayush-56, Ayush-65, the anti-malarial drug and cancer-related research shows that she focused on how chemistry could directly improve people's lives, combining scientific rigor with social responsibility. In other words, she viewed chemistry as a tool for practical human benefit, not just an academic pursuit.

Science in Service of the Nation: A Scientist's Role in Policy and Parliament

Parallel to her laboratory commitments, Prof. Asima, served in national advisory capacities. In 1982 she was nominated to the Rajya Sabha, the Upper House of India's Parliament, by the President of India. Her nomination was recognition not of political ambition but of intellectual authority. In Parliament, she advocated for strengthening scientific research, particularly



(Figure 3.2: Prof. Asima Chatterjee speaking at the Joint Session of Parliament on Birth Centenary Celebration of Dr. Sarvepalli Radhakrishnan 1988. Rights: Prof. Julie Banerji, personal family archives. Credit: Indian Academy of Sciences.)

within universities. She emphasized that research funding must be seen not as expenditure but as investment in national capacity. Her parliamentary tenure illustrated another dimension of her public life: she believed that scientists bore responsibility not only for discovery but also for shaping policy frameworks that support discovery.

Honours and Recognitions:

Prof. Asima's recognition during this period continued. The Padma Bhushan in 1975 had already acknowledged her national stature. Later, she received several additional honours, including the Shanti Swarup Bhatnagar Prize in Chemical Sciences, one of India's highest recognitions for scientific research. Each award represented a different facet of her contribution: research excellence, institutional leadership, and public engagement. Yet those who knew her closely often observed that she wore honours lightly. Students visiting her office encountered not self-display but disciplined conversation. Her desk was orderly; manuscripts were annotated with care; references were meticulously cited. She insisted that scientific writing must be precise, because imprecision in language often reflected imprecision in thought.

Think Like a Physicist – or a Chemist

When evaluating a medicinal plant, one must ask systematic questions. What is the active compound? What is its molecular structure? How does that structure influence interaction with biological targets? What chemical transformations occur within the body? Does the compound degrade? Can it be synthesized or modified? This sequence of questioning mirrors the scientific method itself: observe, hypothesize, test, refine.

The laboratory she led during these decades became a crucible of rigorous training. Many of her doctoral students went on to occupy important academic and research positions across India. They carried with them not only technical skill but an ethos: respect for evidence, patience in experimentation, and intellectual humility before data. Former students often recalled that she rarely raised her voice; instead, she would pose a quiet but penetrating question that compelled deeper thought. A flawed argument was corrected not through rebuke but through reasoning.

Her engagement with international scientific communities continued as well. She participated in conferences abroad, presented research findings, and contributed to authoritative review volumes on alkaloid chemistry. These review articles were particularly significant. They required synthesis of vast literature, critical evaluation of competing structural proposals, and clarity in presentation. Through such contributions, she influenced not only primary research but interpretative frameworks within natural product chemistry.

Even as she advanced in age, her intellectual energy did not diminish. Retirement from formal university service in 1982 did not signify withdrawal from research. She continued to guide projects, advise younger scientists, and contribute to discussions on medicinal chemistry. The boundary between professional and personal life remained porous, shaped by enduring commitment to scholarship.

The deeper significance of this phase of her life lies in the integration it achieved. Structural chemistry, pharmacology, traditional medicine, public health policy, and education were not isolated compartments in her thought. They were interconnected dimensions of a single intellectual vocation. The molecules she studied were not abstract entities; they were potential

From Vision to Reality: Establishment of an Ayurvedic Research Institute

Since the beginning of her teaching and research career, Prof. Asima Chatterjee envisioned establishing an institute dedicated to the study of Indian medicinal plants, development of new ayurvedic formulations, and the creation of an ayurvedic hospital to serve the people of West Bengal. This vision began to take shape when she received a generous donation of 3.5 acres of land in Salt Lake City, Kolkata, from the then Chief Minister, Jyoti Basu. She further secured financial support in the form of a ₹4 crore building grant from the Ministry of Health and Family Welfare, Government of India.

This remarkable collaboration between the state and central governments led to the establishment of the Regional Research Institute (Ayurveda), which was later upgraded to the Central Research Institute (Ayurveda) under the direct administration of the Ministry of Health and Family Welfare. Prof. Chatterjee served as its Honorary Principal Coordinator for many years, playing a pivotal role in its development.

The institute houses an Ayurvedic hospital along with specialized research centers in chemistry, botany, and pharmacology, focusing on drugs derived from Indian medicinal plants. It is actively engaged in developing Ayurvedic formulations, conducting systematic clinical trials, and distributing these medicines across India.

instruments of healing. The laboratories she built were not private spaces; they were national assets. The students she mentored were not merely thesis candidates; they were inheritors of a scientific tradition.

As the 1990s approached, India's scientific landscape was evolving rapidly, shaped by liberalization, technological expansion, and new research priorities. Yet the foundational work accomplished by scholars like Asima Chatterjee remained integral. She had demonstrated that rigorous science could grow from indigenous knowledge systems without sacrificing international standards. She had shown that a woman scientist in twentieth century India could not only participate in but lead national scientific discourse.

This chapter closes with an image that encapsulates her legacy during these years: a chemist standing at the intersection of tradition and modernity,

holding in one hand the analytical tools of structural reasoning and in the other the ethical responsibility of public health. The journey that began with plant alkaloids had matured into a broader engagement with national well-being. Sustained research, when guided by intellectual discipline and public purpose, eventually draws institutional recognition-not as reward, but as acknowledgment of contribution.

The upcoming chapters will turn toward reflection on the facilities available to her in her formative years, her later years, the students she shaped, the honours that marked her path, and the enduring imprint she left upon Indian science. In that concluding movement, the river of her life will widen into the ocean of legacy.

Chapter 4

The Architecture of Legacy: Mentorship, Recognition, and Scientific Values

"The heights by great men reached and kept were not
attained by sudden flight,
but they, while their companions slept,
were toiling upward in the night."
Henry Wadsworth Longfellow

In the later decades of her life, when laboratories she had once entered as a young researcher now bore her intellectual imprint, Asima Chatterjee's presence acquired a different quality. She was no longer only the investigator who deciphered complex alkaloid structures; she had become an institution in herself. Yet this transformation did not distance her from active academic participation. Rather, it deepened her role as mentor, guide, and custodian of scientific standards.

To understand her legacy, one must look beyond publications and awards and enter the quiet spaces of supervision meetings, thesis discussions, and manuscript revisions. Those who trained under her often spoke not of dramatic declarations but of sustained discipline. She believed that scientific character was formed through habit: careful recording of observations, patient repetition of experiments, and precise articulation of conclusions. If a student presented a result without adequate control data, she would gently ask, "How certain are you?" That question, simple in phrasing, contained the essence of scientific responsibility.

Her mentorship extended across decades. Over the course of her career, she supervised numerous doctoral and postdoctoral scholars who later contributed significantly to Indian chemistry and pharmaceutical research.

Her achievements were not accidental; it was cultivated through rigorous training and intellectual independence.

One former student, who later became a professor of organic chemistry at a leading Indian university, recalled that her greatest lesson was not about reaction mechanisms but about integrity. He recounted how she insisted that every structural proposal must be supported by converging evidence. “Speculation,” she would remark, “is permissible only when it invites verification.” This attitude shaped generations of chemists who learned to balance imagination with proof.

Another student, who pursued research in medicinal chemistry within the pharmaceutical industry, described how she encouraged interdisciplinary curiosity. When pharmacological results differed from expected chemical

Mentorship, and Determination Amidst Adversity

Dr. S. C. Prakash, one of Dr. Asima Chatterji's early Ph.D. students, recalls that he witnessed her struggle to establish herself during a time when research facilities were poorly equipped, funding was scarce, and institutional support was minimal. With government organizations like Council of Scientific and Industrial Research (CSIR), Department of Science and Technology (DST), still developing and others yet to emerge, research guides often bore the cost of chemicals, equipment, and almost all spectral analyses to be had from abroad. Students managed with limited or no scholarships, frequently working part-time and covering thesis expenses themselves, despite uncertain career prospects in research.

Dr. S. C. Prakash further mentions that before he joined Dr. Asima Chatterjee, she had a grant of Rs 300/- p.a. and three college teachers as part time research students. As the sole full-time research scholar, he worked with laboratory grant of 1000/- p.a and stipend of Rs.150/- p.m. Basic facilities were lacking, equipment had to be accessed from places like Jadavpur University and Bose Institute, and even solvents were often borrowed from better-equipped labs such as the B. C. Guha laboratory.

During those days, she received strong encouragement from eminent scientists such as Satyendra Nath Bose, Meghnad Saha, S. K. Mitra, B. C. Guha, and Jnan Chandra Ghosh, along with the support of Calcutta University's leadership. Her husband, Baradananda Chatterjee, a noted physical chemist, also stood firmly by her throughout.

predictions, she did not dismiss them. Instead, she urged collaboration with biologists and clinicians. In doing so, she modelled a research culture in which disciplinary boundaries were bridges rather than barriers.

Her laboratory also nurtured women scientists at a time when gender disparities in higher scientific research remained pronounced. Younger women scholars found in her not only a supervisor but a demonstration that intellectual authority and feminine identity need not conflict. She neither dramatized nor minimized the challenges faced by women in science; she addressed them through excellence and steadiness. The presence of a woman chairing departmental meetings, guiding doctoral defences, and speaking at international conferences carried symbolic weight, even when unspoken. Her doctoral students credited her for instilling methodological patience. Years later, they remarked that the patience cultivated in those experiments became the foundation of their success in the field of chemistry. The lesson was clear: foundational rigour enables innovation.

Recognition accompanied her mentorship, yet she approached honours as acknowledgments of collective effort rather than personal triumph.

The Shanti Swarup Bhatnagar Prize in Chemical Sciences, awarded to her in 1961, affirmed her standing among India's foremost researchers. The prize recognized sustained excellence in scientific research and symbolized

A Historic Sibling Achievement: The Remarkable Story of Two Bhatnagar Laureates

Asima Chatterji received the prestigious Shanti Swarup Bhatnagar Prize for Science and Technology in Chemical Science in the year 1961, for her valuable contributions in the field of Chemistry of Natural Products: Development of Drugs from Natural Sources.

Asima's brother Sarashi Ranjan Mookerjee was a renowned surgeon and collaborated with his sister in her research on medicinal plants. Sarashi Ranjan was known for his studies on a number of diseases such as hypertension, hypothermia and epilepsy. He received the Shanti Swarup Bhatnagar Prize for Science and Technology, in 1968, for his contributions to Medical Sciences in the field of Neurobiology; Nuclear Medicine.

Asima and Sarashi Ranjan thus became the only siblings, till date, to have received the most coveted Shanti Swarup Bhatnagar Prize for Science and Technology.

national confidence in her contributions to natural product chemistry. At a time when few women were visible in high level scientific recognition, the award carried quiet transformative significance.

Her election as Fellow of the Indian National Science Academy, in 1960, further marked her integration into the highest circles of Indian scientific deliberation. Fellowship in such academies represents peer recognition of scholarly contribution and intellectual leadership. For younger scientists observing her trajectory, this fellowship reinforced the principle that scholarship rooted in Indian contexts could achieve international respect.

The Padma Bhushan, conferred in 1975, expanded her recognition beyond academic boundaries into the broader national sphere. Civilian honours of this magnitude reflect the state's acknowledgment of contributions to society at large. In her case, the award recognized not only laboratory discoveries but institutional leadership, integration of traditional medicinal knowledge with modern chemistry, and her service in strengthening scientific education.

Her nomination to the Rajya Sabha in 1982 added another dimension to her public profile. Parliamentary nomination for a scientist underscores the belief that scientific reasoning is integral to national governance. In the Upper House, she advocated for enhanced research funding and emphasized the importance of universities as engines of innovation. Her speeches, measured and evidence based, exemplified the same clarity she demanded in laboratory discussions. Even in her later age, she remained intellectually active. Visitors to her residence often encountered a scholar surrounded by books and research papers, engaged in correspondence and reflective reading. Age did not diminish her curiosity. She followed developments in medicinal chemistry and natural product research with sustained interest, occasionally offering thoughtful commentary to former students.

Try to Imagine

Imagine a generation of chemists spread across universities, research institutes, and pharmaceutical laboratories, each carrying forward habits learned in a single laboratory decades earlier. The mentor may not be physically present in each new experiment, yet her influence shapes experimental design, ethical standards, and interpretative caution. Legacy in science is rarely loud; it is cumulative.



(Figure 4.1: Prof. Asima receiving the University Grants' Commission (UGC) CV Raman Award, from Dr. D. S. Kothari. Rights: Prof. Julie Banerji, personal family archives. Credit: Indian Academy of Sciences.)

As the twentieth century drew toward its close, India's scientific landscape became increasingly globalized. Advanced instrumentation became more accessible; computational chemistry expanded; pharmaceutical industries grew. Yet the foundations laid by pioneers of natural product chemistry remained vital. The careful cataloguing of plant-derived compounds, the methodological discipline of structural elucidation, and the insistence on empirical verification continued to inform contemporary research.

Asima Chatterjee passed away on 22 November 2006, leaving behind not merely publications and honours but a living network of scholars. Obituaries and tributes emphasized her pioneering status as one of India's earliest women doctoral scientists and her leadership in natural product chemistry. Colleagues described her as firm yet compassionate, intellectually demanding yet personally considerate. The language of remembrance often returned to a single theme: integrity.

Her life invites reflection on the nature of scientific legacy. Discoveries may eventually be revised; structures may be reinterpreted with new tools; formulations may be improved. Yet the deeper inheritance lies in intellectual character; the disciplined curiosity, ethical steadfastness, and commitment to public good that she exemplified.

As the story of Prof. Asima Chatterjee reaches conclusion, one senses that her story does not end with biographical chronology. It extends into questions of value:

- What does it mean to integrate tradition with modern science responsibly?
- How can mentorship shape national research culture?
- In what ways can a scientist participate in public life without compromising scholarly depth?

Awards and Honours of Asima Chatterjee

Prof. Asima Chatterjee received multiple prestigious awards and honours in recognition of her significant contributions.

- Sir P. C. Ray Award (1974) and Professor P. K. Bose (1988) Lectureships of the Indian Chemical Society
- Received an honorary D.Sc. degree from the University of Burdwan (1976)
- As a national delegate, spoke on "Women and Work" at the World Congress of Women in Prague (1981)
- Received the Bhuban Mohini Das Gold Medal from the University of Calcutta for her work "Bharater Bonousadhi," in 1981
- Sir C. V. Raman Award of the Hari Om Ashram Trust (1982) by the University Grants Commission
- Received an honorary D.Sc. from Banaras Hindu University (1982)
- Professor Sisir Kumar Mitra Lectureship (1984) of the Indian National Science Academy
- Sir Asutosh Mookerjee Gold Medal (1989) and Dr. G. P. Chatterjee Lectureship (1994) of the Indian Science Congress Association
- Goyal Award (1992) by the Goyal Foundation, Kurukshetra University
- The Indira Gandhi Priyadarshini Award, All India Unit Conference (1994)
- The Silver Jubilee Award (1995) of the Central Council for Research in Ayurveda and Siddha (Government of India)
- The Eminent Teacher Award (1997) by Calcutta University,
- The Rathindra Award of the Visva Bharati University (1997).

As her achievements gained acknowledgment, her influence widened, and the laboratory became a nationally significant platform. The question now shifted from discovery to direction: how might scientific culture itself be strengthened?

The final chapters will draw these threads together, reflecting upon her enduring contribution to Indian science and exploring how her life continues to offer guidance to future generations. There, the river of biography will widen into a meditation on continuity-how knowledge, once cultivated with care, outlives the hands that first shaped it. Her philosophy in life was imbibed with a strong work ethic.

Chapter 5

Natural Product Chemistry in the Mid-Twentieth Century: Context and Methodology

*"The woods are lovely, dark and deep,
But I have promises to keep,
And miles to go before I sleep."*

Robert Frost

The development of natural product chemistry during the mid-twentieth century represents one of the most intellectually demanding phases in the history of organic chemistry. Researchers working in this field faced an unusual combination of complexity and limitation: plant-derived compounds exhibited structural intricacy, yet analytical technologies were comparatively modest. The career of Asima Chatterjee unfolded within this methodological environment, and her scientific discipline cannot be fully appreciated without understanding the techniques available at the time.

1. Conceptual Foundations of Natural Product Chemistry

Natural product chemistry is associated with the isolation, purification, structural determination, and functional evaluation of chemical compounds produced by living organisms. In plant systems, such compounds often serve ecological roles such as defence against herbivores, microbial resistance, or signalling. Many of these molecules exhibit pharmacological activity in humans.

The intellectual challenge of the discipline lies in its starting point: a complex biological matrix rather than a defined synthetic precursor. A chemist does not begin with a known molecular formula but with plant material containing a multitude of constituents. The task is therefore investigative rather than - just synthesis.

Protonation

Protonation is defined as the addition of a proton (H^+) to a molecule, which often occurs in the first step of nucleophilic addition or substitution reactions involving carbonyl groups, resulting in protonated intermediates that can stabilise the reaction.

During the 1940s through the 1960s, structural organic chemistry had already established theoretical frameworks- valency, resonance, stereochemistry, and reaction mechanisms- but practical structural elucidation depended upon incremental experimental deduction.

2. Extraction Techniques : Acid–Base Partitioning

One of the most important early steps in alkaloid research was acid–base extraction. Alkaloids, being nitrogen-containing bases, form salts in acidic environments. When plant material is treated with dilute acid, alkaloids dissolve in the aqueous phase as protonated species. Non-basic organic compounds remain in organic solvents.

Subsequent basification of the aqueous solution liberates the free alkaloids, which can then be extracted into an organic solvent such as chloroform or ether. This selective solubility forms the foundational separation strategy.

The essence of acid–base partitioning lies in its chemical logic: exploiting intrinsic molecular properties to achieve separation without sophisticated instrumentation. Mastery of such techniques required both theoretical understanding and practical precision.

3. Fractional Crystallization and Chromatography

After initial extraction, mixtures of alkaloids often required further separation. Fractional crystallization utilized differences in solubility across temperature gradients. Repeated crystallization gradually enriched individual compounds.

Column chromatography, increasingly adopted during the mid-twentieth century, provided improved resolution. Adsorbents such as alumina or silica gel allowed differential migration of compounds based on polarity. Although modern high-performance liquid chromatography did not yet exist, careful optimization of solvent systems allowed effective separation.

Chromatographic visualization methods, such as Dragendorff's reagent for alkaloids, provided qualitative confirmation. These techniques demanded patience and empirical refinement.

4. Elemental Analysis and Molecular Formula Determination

Combustion analysis determined the percentages of carbon, hydrogen, and nitrogen in a compound. From these percentages, empirical formulae could be calculated. Molecular weight determination, through methods such as cryoscopy or mass measurement techniques available at the time, allowed the establishment of molecular formula.

The combination of empirical formula and degree of unsaturation provided the first structural constraints. For example, high unsaturation might indicate aromatic rings or multiple double bonds. Such numerical reasoning formed the backbone of structural deduction.

5. Functional Group Identification

Infrared spectroscopy, gradually introduced into Indian laboratories during the 1950s and 1960s, allowed identification of characteristic vibrational frequencies. Carbonyl groups, hydroxyl groups, amines, and aromatic systems displayed distinct absorption bands.

Ultraviolet-visible spectroscopy provided insight into conjugated systems. Indole alkaloids, for instance, exhibited characteristic UV absorption due to their aromatic frameworks.

However, spectral data did not automatically reveal complete structure. Interpretation required correlation with chemical reactivity.

6. Chemical Degradation Studies

Before widespread access to high-resolution nuclear magnetic resonance spectroscopy, structural elucidation relied heavily upon chemical transformation. By subjecting a compound to controlled reactions like oxidation, reduction, hydrolysis, chemists generated simpler fragments.

If oxidation yielded a specific acid derivative known from previous literature, the presence of a corresponding side chain could be inferred. If hydrolysis produced an identifiable amine, ring connectivity could be deduced. Stepwise degradation functioned as logical disassembly.

This methodology required extraordinary caution. Side reactions could mislead interpretation. Therefore, reproducibility and confirmation through multiple independent reactions were essential.

7. Optical Rotation and Stereochemistry

Many natural products are chiral. Measurement of optical rotation provided insight into stereochemical configuration. While absolute configuration often remained uncertain without X-ray crystallography, relative stereochemistry could sometimes be deduced through comparative studies.

Understanding stereochemistry is crucial because biological systems are stereospecific. Two enantiomers of the same molecular formula can exhibit dramatically different biological activities. Thus, stereochemical reasoning linked structural chemistry to pharmacological relevance.

8. Emergence of Nuclear Magnetic Resonance (NMR)

By the late 1960s and 1970s, nuclear magnetic resonance spectroscopy began transforming structural analysis. Proton NMR revealed information about hydrogen environments, coupling patterns, and functional group adjacency. Carbon-13 NMR later expanded structural clarity.

For chemists trained in classical methods, NMR represented both liberation and responsibility. Spectral data accelerated structural confirmation but required disciplined interpretation. Those trained under rigorous deductive traditions were well positioned to integrate these advances without over reliance.

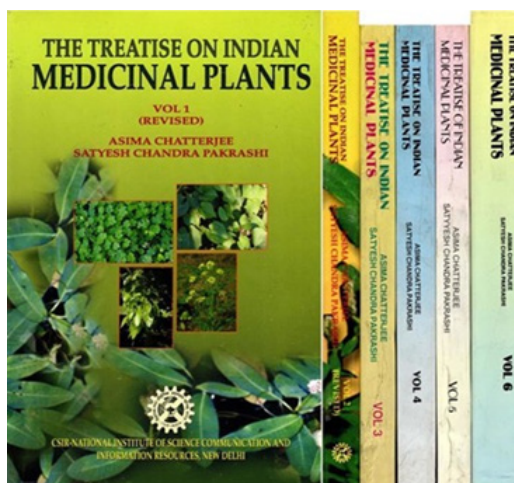
9. Structure–Activity Relationship (SAR)

As medicinal implications became central, the concept of structure–activity relationship gained importance. SAR analysis seeks to correlate molecular modifications with biological response. For example, introducing an electron-withdrawing group may enhance binding affinity to a biological target; altering ring size may influence membrane permeability.

SAR reasoning bridges organic chemistry and pharmacology. It transforms structural knowledge into predictive capacity. In laboratories focused on plant-derived compounds, SAR analysis often emerged through comparative study of closely related alkaloids.

10. Phytochemical Standardization

When research progressed from isolated compounds to multi-component herbal formulations, standardization became scientifically essential. Variability in plant chemistry can arise from soil composition, climate, harvesting season, and processing method. Therefore, establishing chemical markers—compounds whose concentration can be quantified—ensures batch consistency.



(Figure 5.1: Treatise on Medicinal Plants)

Prof. Asima Chatterji was the Chief Editor of the six volumes of Treatise on Medicinal Plants, which described seven hundred medicinal plants and was published by the CSIR-National Institute of Science Communication and Information (CSIR-NISCAIR)

Chromatographic fingerprinting became a crucial tool. By comparing retention patterns across batches, chemists confirmed chemical similarity. Quantitative assays provided dosage control.

Standardization is not merely technical; it is ethical. In medicinal contexts, consistency helps to build and maintain patients' trust.

11. Interdisciplinary Integration

Natural product chemistry integrates botany, pharmacology, toxicology, and clinical medicine. Plant identification must be taxonomically accurate. Biological assays must follow validated protocols. Toxicological evaluation must precede clinical use.

The chemist's role is foundational: without structural clarity and purity assurance, biological interpretation remains uncertain. The integration of these disciplines reflects the maturation of medicinal chemistry.

12. Intellectual Discipline as Method

Beyond technique, mid-twentieth century natural product chemistry demanded intellectual virtues: patience, scepticism, precision, and resilience. Instrumentation was limited; error margins were significant. Only through repeated confirmation could confidence be achieved.

The generation trained during this period developed an unusually strong foundation in deductive reasoning. When modern instrumentation later expanded analytical power, those foundations ensured responsible interpretation.

The scientific methods described above formed the methodological landscape within which pioneering chemists in India built their research traditions. Their achievements cannot be evaluated solely by contemporary technological standards. Instead, they must be understood as products of disciplined reasoning applied under constraint.

Natural product chemistry during this period exemplified a union of empirical tradition and modern analytical rigor. It required respect for biological diversity and commitment to molecular precision. The legacy of such work continues to enhance contemporary drug discovery and phytochemical research.

Laboratory Culture and Scientific Methodology: A Legacy of Prof. Asima Chatterjee

The scientific contributions of a laboratory cannot be separated from its internal culture. The laboratory of Prof. Asima Chatterjee was characterized by meticulous record-keeping. Laboratory notebooks were not informal diaries but formal scientific documents. Data entries were dated, procedures described in detail, and deviations carefully noted. Students learned that future reproducibility depended upon present clarity.

Instrumental techniques gradually became more accessible in India during the later decades of her career. Infrared spectroscopy, ultraviolet-visible spectroscopy, and later nuclear magnetic resonance spectroscopy enriched structural analysis. Yet she ensured that students first mastered classical reasoning. Instrumental output, she insisted, must be interpreted critically. A spectrum does not interpret itself; it must be understood in context.

Her insistence on methodological grounding proved invaluable as Indian laboratories modernized. Students trained in logical deduction adapted readily to new technologies because they understood underlying principles rather than merely operating instruments.

Science and Ethical Responsibility

The expansion of her work into medicinal applications brought ethical considerations to the forefront. When dealing with compounds that affect human health, precision is not merely academic; it is moral. Overstatement

of results, inadequate testing, or premature claims could have serious consequences. Her careful language in publications reflected this awareness. Conclusions were stated proportionally to evidence.

In parliamentary discussions and academic addresses, she often emphasized that scientific progress must be accompanied by responsibility. Universities, she argued, must foster not only technical competence but ethical awareness. This principle resonated with her own career trajectory, which linked laboratory research to public health and policy.

Integration of Indigenous Knowledge and Modern Chemistry

Perhaps the most enduring scientific theme of Asima Chatterji's career is integration. India's vast medicinal plant heritage represents accumulated empirical knowledge. However, empirical observation alone does not satisfy modern scientific criteria. By applying structural chemistry to these plants, she demonstrated that indigenous knowledge could be translated into globally intelligible chemical language.

This translation did not reduce or dismiss tradition; rather, it expressed it more clearly. Chemical structures do not erase cultural history; they clarify mechanisms. In doing so, she contributed to a uniquely Indian model of scientific modernization, rooted yet rigorous.

Toward a Fuller Scientific Assessment

When evaluating Asima Chatterji's scientific legacy, several dimensions stand out:

- First, her contributions to indole alkaloid chemistry enriched international structural understanding.
- Second, her work on coumarins and related compounds strengthened medicinal chemistry research in India.
- Third, her leadership in anti-malarial research bridged laboratory chemistry and therapeutic application.
- Fourth, her mentorship institutionalized methodological discipline across generations.

These dimensions collectively moved Indian natural product chemistry from exploratory beginnings toward internationally respected scholarship.

Noteable Contributions

Among Asima Chatterji's contributions, a few notable ones are:

- Initiated chemical investigation of alkaloids in *Rauwolfia canescens*.

- Investigated the chemistry of almost all principal types of indole alkaloids.
- Contributions with regard to elucidation of structure and stereochemistry of ajmalicine and sarpagine.
- First suggested stereo-configuration of sarpagine.
- Isolated and characterized geissoschizine, a key precursor in biogenesis of indole alkaloids from *Rhazya stricta*.
- Carried out synthetic studies on a number of complex indole alkaloids, quinoline and isoquinoline alkaloids.
- Developed simplified and normal procedure for preparation of required beta phenylethanolamines in connection with alkaloid synthesis.
- Elucidated the structure of luvangetin isolated from *Luvanga scandens*.
- Studied the action of various Lewis' acids on prenylated coumarins and devised simple synthetic routes to a number of complex coumarin systems.
- Investigated mechanism of acid-catalysed hydramine fission of beta phenylethanol amines.
- Introduced use of periodic acid as a reagent for detection and location of both terminal and exocyclic double bonds in organic compounds.

Chapter 6

Science, Service, and Ethical Outlook of a Chemist

"Lives of great men all remind us, we can make our lives sublime,
and, departing, leave behind us, footprints on the sands of time."

Henry Wadsworth Longfellow

Institutions endure; individuals pass. Yet the arc of a scientific life offers a lens through which institutional transformation becomes focused and attains a creditable status.. In the long arc of scientific history, individual lives are often remembered for discoveries, for formulas that enter textbooks,



(Figure 6.1: Google Doodle Honouring Asima Chatterjee on her 100th Birthday, 23 September 2017, Image credit: google.com)

Google commemorated Indian chemist Asima Chatterjee with her very own Google Doodle. The doodle shows a smiling Asima Chatterjee coloured in green and the Google logo designed to resemble chemical bonds.

or for institutions that bear a scholar's name. Yet some lives invite a different kind of remembrance. They prompt reflection not only on what was discovered, but on how knowledge was pursued and why it was valued. The life of Asima Chatterjee belongs to this latter category. Her work in natural product chemistry, her contributions to medicinal research, her leadership in academia, and her service to national institutions form a coherent whole because they were guided by a consistent intellectual ethic.

From the earliest stages of her career, she demonstrated that scientific inquiry in India need not be derivative imitation of Western paradigms. Instead, it could draw confidently from indigenous botanical knowledge while applying the most rigorous analytical tools available. This orientation required balance. Traditional medicinal practices could not be idealized; they required experimental validation. At the same time, modern chemistry could not afford cultural amnesia; it had to recognize the empirical insights embedded in centuries of observation. Her work on plant alkaloids and anti-malarial formulations exemplified this equilibrium.

Asima Chatterji's intellectual courage was quiet rather than dramatic. When resources were scarce, she relied on disciplined reasoning. When confronted with institutional limitations, she built laboratories. When personal tragedy struck, she returned to work not as retreat but as advancing purposefully. Her nomination to the Rajya Sabha reflected a conviction that scientific thinking has relevance beyond laboratories. Her speeches there echoed the same theme that had guided her academic life: universities are not peripheral; they are central to national development.

In assessing her contributions, one must consider the historical context in which she worked. The first half of the twentieth century offered limited institutional support for women in advanced scientific research. Laboratories were predominantly male spaces; academic hierarchies were rigid; expectations of domestic responsibility were heavy. That she not only entered but led such spaces marks a transformation in Indian scientific culture. Yet she did not present her work in adversarial terms. Her authority rested on knowledge. Her presence altered expectations through demonstration rather than proclamation.

The moral imagination of a scientist, as embodied in her life, lies in the capacity to see knowledge as service. The molecules she studied were not abstractions; they were potential remedies. The students she trained were not mere apprentices; they were future custodians of research culture. The awards she received were not endpoints; they were responsibilities to sustain

standards. Her scientific publications enriched international literature, but her institutional work strengthened national capacity.

The literary and editorial contributions of Prof. Asima Chatterjee are also immense in terms of the volume of the contributions done by her. For example,

- On the request of the late Professor Satyendra Nath Bose, FRS, she wrote *Sarai Madhyamic Rasayan*, a book in Bengali on chemistry for secondary school students, published by Bangiya Bijnan Parishad, an Institute for the Popularisation of Science founded by SN Bose himself.
- Prof. Chatterjee edited and revised the six-volume *Bharatiya Banoushodhi* (originally compiled by the late Dr. K. P. Biswas), a treatise in Bengali on Indian Medicinal Plants in six volumes (Volumes 1-5; 1973; Volume 6; 1977) and published by the Calcutta University Press.
- She was also the Chief-Editor of the six-volume series, *The Treatise of Indian Medicinal Plants*, published by CSIR.

Prof. Asima Chatterjee dedicated her whole life to carry out landmark researches in the field of medicinal chemistry. Also, her devoted work in the field of science have paved the way for millions of women to excel in this field. Her philosophy in life was imbibed in a very strong work ethic, *"I wish to work as long as I live."* Her passing in November 2006, at the age of 89, marked the end of a life that had spanned colonial rule, independence, post-independence reconstruction, and the early decades of globalized science. Few scientists experience such sweeping historical transitions. Through them all, she remained anchored in disciplined inquiry.

Other Awards and Honours of Prof. Asima Chatterjee

Prof. Asima Chatterjee received multiple prestigious awards and honours in recognition of her significant contributions.

- Awarded the Academy Medal by the West Bengal Academy of Science and Technology (1998),
- Received the Sir Devaprasad Sarbadhikary Gold Medal, the most prestigious award of the University of Calcutta (1999),
- Received an honorary D.Sc. from the University of Kalyani (1999),
- Received the PC Chandra Purshkar from the PC Chandra Group (2001),
- Received an honorary D.Sc. from the Vidyasagar University (2006).

To read her life today is to encounter a model of integration. Learning and knowledge need not detach itself from society; it may illuminate it. Tradition need not be rejected to embrace modernity; it may be reinterpreted through critical method. Authority need not be loud to be transformative; it may rest in patient consistency.

The river of her life, which began in a household attentive to books and ideas, widened through classrooms, laboratories, parliamentary halls, and international conferences. It entered the ocean of collective memory not as a solitary current but as part of a larger flow of Indian scientific endeavour. In that ocean, her name endures as a symbol of disciplined curiosity and public commitment.



(Figure 6.2: Prof. Asima Chatterjee with her students on her Eightieth Birth Anniversary at Science City 23rd September 1997. Rights: Prof. Julie Banerji, personal family archives. Credit: Indian Academy of Sciences.)

Appendix – A: Chronological Timeline of Major Milestones

Year	Event
1917	Born Asima Mukherjee in Calcutta (now Kolkata) on September 23.
1932	Passed Matriculation Examination from Bethune Collegiate School, Kolkata, securing a Bengal Government Scholarship.
1933	Stood second in the All-Bengal Music Competition.
1934	Passed the Intermediate in Science (ISc) Examination from Bethune College; received several scholarships and the Hemprova Bose Memorial Medal.
1936	Graduated with honours in Chemistry from Scottish Church College, receiving the Basanti Das Gold Medal. She was the only woman among three admitted to complete the programme.
1938	Obtained her M.Sc. degree in Organic Chemistry from the University of Calcutta, receiving a silver medal and the Jogmaya Devi Gold Medal. Began research work under Dr. Prafulla Kumar Bose on natural products.
1940	Received the Nagarjuna Prize and gold medal from the University of Calcutta for her research. Joined Lady Brabourne College as the Founder-Head of the Department of Chemistry.
1942	Received the Premchand Roychand Studentship.
1944	Awarded the Mouat Gold Medal and a DSc degree from the University of Calcutta based on her research on indole alkaloids and coumarins, becoming the first woman to receive a DSc from any Indian university. Appointed Honorary Lecturer in the Department of Chemistry, University of Calcutta.
1945	Married Dr. Baradananda Chatterjee.
1947	Left for the USA on study leave with her infant daughter. Worked with Professor L.M. Parks at the University of Wisconsin on naturally occurring glycosides (1947-1948).
1948	Worked with Professor L. Zechmeister at the California Institute of Technology on carotenoids and provitamin A (1948-1949); awarded the Watumull Fellowship for this work.
1949	Worked with Professor Paul Karrer at the University of Zürich, Switzerland, on biologically active indole alkaloids (1949-1950), which became a lifelong interest.
1950	Returned to India and rejoined Lady Brabourne College and the University of Calcutta, starting intensive research on alkaloids and coumarins.
1954	Appointed Reader in the Department of Chemistry, University of Calcutta.
1960	Elected a Fellow of the National Institute of Sciences of India (now the Indian National Science Academy, INSA).
1961	Awarded the Shanti Swarup Bhatnagar Award for Chemical Sciences by the CSIR, Government of India.

1962	Became the Kumar Guruprasad Singh Khaira Professor of Chemistry at the University of Calcutta, a chair she held until her retirement in 1982.
1967	(April-September) Suffered a personal tragedy with the death of her father and then her husband within four months. Subsequently suffered a massive heart attack.
1968	Served as the principal witness for Bengal Chemical and Pharmaceutical Works Ltd. in a historic patent infringement legal battle against Hoechst Co. Ltd. in the Calcutta High Court.
1969	Became the Head of the Department of Chemistry, University of Calcutta, serving until 1979.
1972	Became the Honorary Programme-Coordinator for the University Grants Commission's Programme of Special Assistance for intensifying teaching and research on Natural Product Chemistry.
1975	Elected General President of the 62 nd Session of the Indian Science Congress in Delhi, the first woman scientist to hold this position. Conferred the Padma Bhushan by the Government of India.
1976	Received an honorary D.Sc. degree from the University of Burdwan. She also patented the anti-epileptic drug combination named Ayush-56 (Patent No. 141170, July 14, 1976).
1979	Patented the anti-malarial drug combination named Ayush-64 (Patent No. 568/Del/70, August 7, 1979).
1981	As a national delegate, spoke on "Women and Work" at the World Congress of Women in Prague. Received the Bhuvan Mohini Das Gold Medal from the University of Calcutta for her work "Bharater Bonousadhi."
1982	Awarded the Sir C.V. Raman Award by the University Grants Commission. Retired from the Khaira Professorship in Chemistry. Nominated by the President of India as a Member of Parliament (Rajya Sabha), as a Scientist-Academician (February 1982- April 1984; May 1984 - May, 1990). Received an honorary D.Sc. from Banaras Hindu University.
1984	Received the Sisirkumar Mitra Lectureship of the Indian National Science Academy.
1989	Received the Sir Asutosh Mukherjee Memorial Gold Medal, the most prestigious award of the Indian Science Congress Association.
1997	Received the Eminent Teacher Award from the University of Calcutta.
1998	Awarded the Academy Medal by the West Bengal Academy of Science and Technology.
1999	Received the Sir Devaprasad Sarbadhikary Gold Medal, the most prestigious award of the University of Calcutta. Received an honorary D.Sc. from the University of Kalyani.
2001	Received the PC Chandra Purshkar from the PC Chandra Group.
2006	Passed away on November 22, 2006.
2017	Google honoured her with a Doodle on her 100th birth anniversary.

Appendix – B:

Major Publications

Prof. Asima Chatterjee has published more than 300 research publications in the field of her work. A few of the prominent of her scientific publications are:

Title	Year
The treatise on Indian medicinal plants: vol. 1 A Chatterjee, SC Pakrashi New Delhi: Publications and Information Directorate, CSIR 172p.-illus., col ...	1991
The reversible antifertility effect of <i>Piper betle</i> Linn. on Swiss albino male mice M Sarkar, P Gangopadhyay, B Basak, K Chakrabarty, J Banerji, ... Contraception 62 (5), 271-274	2000
Lewis-acid-induced electrophilic substitution in indoles with acetone. Part 2 A Chatterjee, S Manna, J Banerji, C Pascard, T Prangé, JN Shoolery Journal of the Chemical Society, Perkin Transactions 1, 553-555	1980
Occurrence of geissoschizine and other minor biogenetically related alkaloids in <i>Rhazya stricta</i> A Banerji, PL Majumder, A Chatterjee Phytochemistry 9 (7), 1491-1493	1970
Strictamine HK Schnoes, K Biemann, J Mokry, I Kompis, A Chatterjee, G Ganguli The Journal of Organic Chemistry 31 (5), 1641-1642	1966
Gadain, a lignan from <i>Jatropha gossypifolia</i> J Banerji, B Das, A Chatterjee, JN Shoolery Phytochemistry 23 (10), 2323-2327	1984
On the Constitution of the Active Principles Isolated from the Matured Bark of <i>Aegle marmelos</i> , Correá A Chatterjee, SS Mitra Journal of the American Chemical Society 71 (2), 606-609	1949

Title	Year
Structure and Stereochemistry of Nardostachysin, a New Terpenoid Ester Constituent of the Rhizomes of <i>Nardostachys jatamansi</i> A Chatterjee, B Basak, M Saha, U Dutta, C Mukhopadhyay, J Banerji, ... Journal of natural products 63 (11), 1531-1533	2000
LEWIS ACID INDUCED ELECTROPHILIC SUBSTITUTION OF INDOLE. PART 3 J Banerji, A Chatterjee, S Manna, C Pascard, T Prangé, JN Shoolery ChemInform 12 (21)	1981
Alstonia scholaris: Struktur des Indolalkaloides Narelin Y Morita, M Hesse, H Schmid, A Banerji, J Banerji, A Chatterjee, ... Helvetica chimica acta 60 (4), 1419-1434	1977
Saradaferin, a new sesquiterpenoid coumarin from <i>Ferula assafoetida</i> D Bandyopadhyay, B Basak, A Chatterjee, TK Lai, A Banerji, J Banerji, ... Natural product research 20 (10), 961-965	2006
Antifertility effect of Piper betle Linn. extract on ovary and testis of albino rats. P Adhikary, J Banerji, D Chowdhury, AK Das, CC Deb, SR Mukherjee, ... Indian journal of experimental biology 27 (10), 868-870	1989
Studies on the chemical constituents of <i>Nardostachys jatamansi</i> DC (Valerianaceae) A Chatterjee, B Basak, U Datta, J Banerji, A Neuman, T Prange CSIR	2005
Coumarins of <i>Murraya exotica</i> —absolute configuration of auraptenol BR Barik, AK Dey, PC Das, A Chatterjee, JN Shoolery Phytochemistry 22 (3), 792-794	1983
Alkaloidal constituents of <i>eganum harmala</i> and synthesis of the minor alkaloid deoxyvascinone A Chatterjee, M Ganguly Phytochemistry 7 (2), 307-311	1968
Extracts of garlic as possible source of insecticides [<i>Allium sativum</i> Linn., India]. D Kirtaniya, MR Ghosh, N Adityachaudhury, A Chatterjee Indian Journal of Agricultural Sciences	1980

Title	Year
Extracts of garlic as possible source of insecticides. S Deb-Kirtaniya, MR Ghosh, N Adityachaudhury, A Chatterjee Indian Journal of Agricultural Sciences 50 (6), 507-510	1980
NEW COUMARINO-LIGNOIDS FROM HEMIDESMUS INDICUS R. BR. PC Das, PC Joshi, S MANDAL, DAS ASESH, A Chatterjee, A Banerji Indian journal of chemistry. Sect. B: Organic chemistry, including medical ...	1992
Cleogynol, a Novel Dammarane Triterpenoid from <i>Cleome gynandra</i> PC Das, A Patra, S Mandal, B Mallick, A Das, A Chatterjee Journal of natural products 62 (4), 616-618	1999
Crotocaudin; a rearranged labdane type norditerpene from <i>Croton</i> <i>caudatus</i> geisel A Chatterjee, A Banerjee, F Bohlmann Tetrahedron 33 (18), 2407-2414	1977

Appendix – C: Further Reading and Exploration

All those interested in gaining a deeper appreciation of the life and enduring contributions of Prof. Asima Chatterjee may refer to the following carefully selected websites, digital archives, and scholarly works. Together, these resources shed light on her intellectual pursuits, spirit of collaboration, and lasting influence on scientific research, offering valuable perspectives for both researchers and general readers.

- Indian National Science Academy (INSA). Fellowship records and biographical notices.
- Indian Academy of Sciences: https://www.ias.ac.in/public/Resources/Initiatives/Women_in_Science/Contributors/Chatterjee.pdf
- Proceedings of the Indian Science Congress Association, Presidential Address (1975).
- Asima Chatterjee's research works, University of Calcutta
<https://scholar.google.com/citations?user=Ww7HMPQAAAAJ&hl=en>
- Rajya Sabha Secretariat. Parliamentary debates and nominated member records (1982–1988 tenure).
- Pelletier, S. W. (Ed.). *Alkaloids: Chemical and Biological Perspectives*.
- Dewick, P. M. *Medicinal Natural Products: A Biosynthetic Approach*.
- Silverman, R. B. *The Organic Chemistry of Drug Design and Drug Action*.
- Mitra, A. K. *Asima Chatterjee, One of the Brightest Stars in the Galaxy of Organic Chemistry*. <https://www.jstor.org/stable/27310734>.
- <https://artsandculture.google.com/asset/prof-asima-chatterjee/NAGU753ImLWpBw?hl=en>
- <https://paulingblog.wordpress.com/tag/asima-chatterjee/>
- S C Pakrashi. Asima Chatterjee
- De, Asish (1 January 2015). "Asima Chatterjee: A unique natural products chemist". *Resonance: Journal of Science Education*. 20 (1): 6–22. doi:10.1007/s12045-015-0148-9
- Ghosal, Shibnath (8 April 2003). "A Tribute to Prof. Asima Chatterjee". *Arkivoc*. 2003 (9): 1–3. doi:10.3998/ark.5550190.0004.90

ABOUT THE BOOK

The book, *Asima Chatterjee: Unlocking Nature's Chemistry*, offers a comprehensive and nuanced exploration of the life and work of Professor Asima Chatterjee, a pioneer in natural product chemistry and medicinal research in India. Through a detailed narrative, the book examines her pioneering contributions to plant-based medicinal chemistry, her role in bridging traditional knowledge with modern scientific methods, and her lasting impact on academic institutions and public health. It also reflects on her intellectual discipline, resilience, and commitment to knowledge as a form of service. Designed for students, educators, and researchers alike, the book invites readers to engage with science not merely as a discipline, but as a socially responsive and culturally rooted endeavour.

ABOUT THE AUTHORS

K. Venkataraman is former Curator of the National Council of Science Museums, (Govt. of India), Kolkata, and presently a faculty of Physics at PMB Gujarati Science College, Indore. He is an academician, writer and the Editor-in-Chief of Vidyarthi Vigyan Manthan (VVM).

Amritanshu Vajpayee is presently the Principal Investigator of Astro STREAM Project, Saptarishi India (under Global Sky Partners Programme of Las Cumbres Observatory) and Director of S.K.Y. Foundation Lucknow, Uttar Pradesh. He is also an amateur astronomer, author, science communicator, STEAM educator and Co-Editor of VVM.

Nandini Phanse is presently the Head of the Post Graduate Department of Microbiology of PMB Gujarati Science College, Indore. An academician and researcher for 34 years, she is a writer and the Co-Editor of VVM.

Shilpa Parikh is presently the Head of the Computer Science Department in MKHS Gujarati Girls College, Indore. An academician for the last four decades, she is a writer, philanthropist and spiritualist.

Brajesh Pandey is presently the Executive Director at the Indian National Science Academy (INSA), New Delhi. With extensive teaching experience in physics his research areas include nanotechnology, magnetism and the hyperfine structure of materials. He is the Controller of Examination for VVM.

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VIJNANA BHARATI

D-12, South Extension-I, New Delhi- 110049

E-mail: vijnanabharati@gmail.com

Contact: +91-011- 41040846